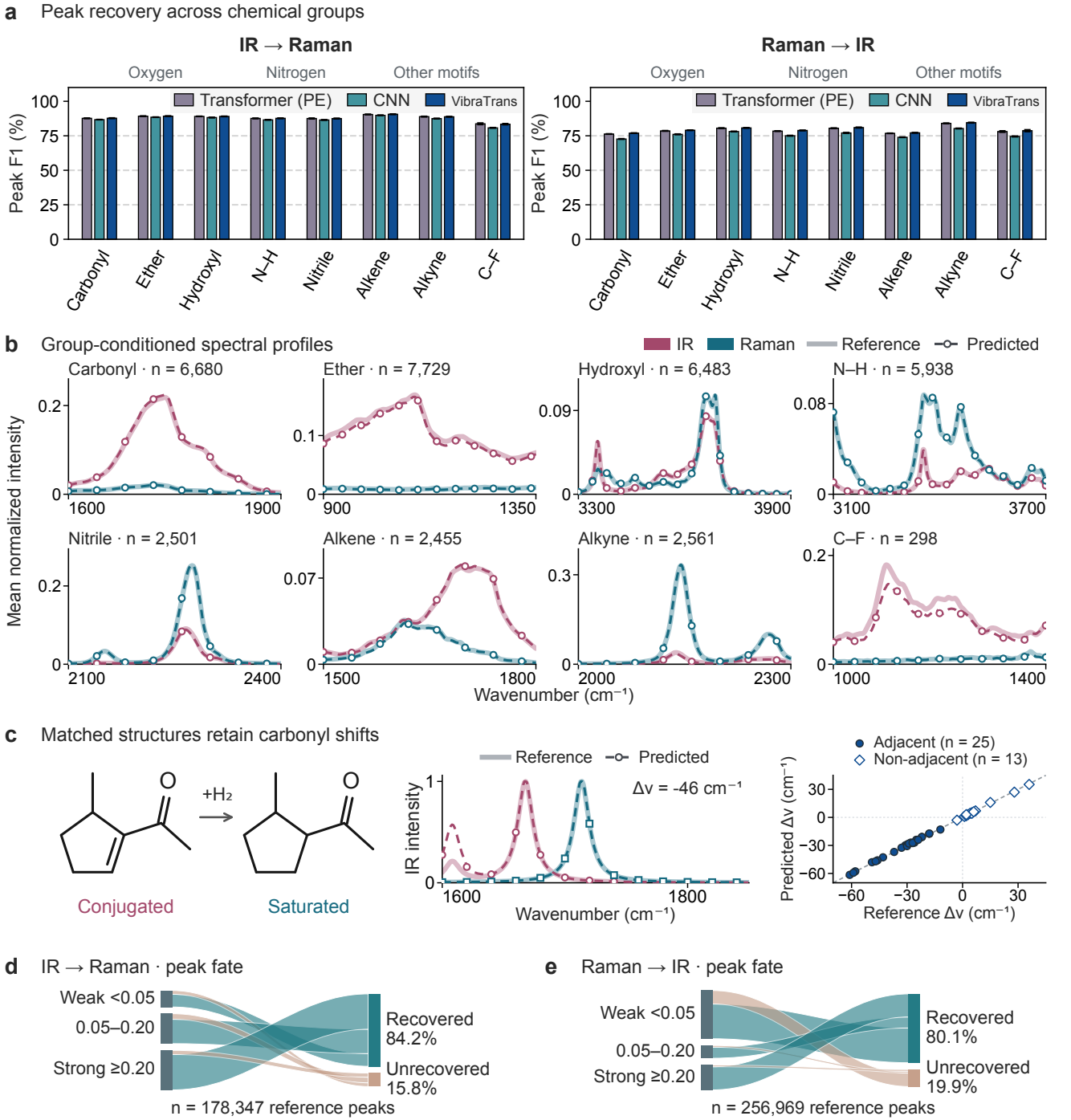

a
Peak recovery across chemical groups
IR → Raman
Raman → IR
Oxygen
Nitrogen
Other motifs
Transformer (PE)
CNN
VibraTrans
Peak F1 (%)
Carbonyl
Ether
Hydroxyl
N–H
Nitrile
Alkene
Alkyne
C–F
b
Group-conditioned spectral profiles
IR
Raman
Reference
Predicted
Carbonyl · n = 6,680
Ether · n = 7,729
Hydroxyl · n = 6,483
N–H · n = 5,938
Nitrile · n = 2,501
Alkene · n = 2,455
Alkyne · n = 2,561
C–F · n = 298
Mean normalized intensity
Wavenumber (cm⁻¹)
c
Matched structures retain carbonyl shifts
+H₂
Conjugated
Saturated
Reference
Predicted
Δν = -46 cm⁻¹
IR intensity
Wavenumber (cm⁻¹)
Adjacent (n = 25)
Non-adjacent (n = 13)
Predicted Δν (cm⁻¹)
Reference Δν (cm⁻¹)
d
IR → Raman · peak fate
Weak <0.05
0.05–0.20
Strong ≥0.20
Recovered
84.2%
Unrecovered
15.8%
n = 178,347 reference peaks
e
Raman → IR · peak fate
Weak <0.05
0.05–0.20
Strong ≥0.20
Recovered
80.1%
Unrecovered
19.9%
n = 256,969 reference peaks